

RECRUITAI

PROJECT SPECIFICATION & IMPLEMENTATION BLUEPRINT

Fullstack Task Management Dashboard

Detailed project brief generated from the project record.

CANDIDATE

Varun Chopra

DURATION

14 days

TECHNOLOGY STACK

Angular

TypeScript

Java

Spring Boot

MySQL

REST API

Project Overview

Summary of the problem, purpose and expected outcome.

PROJECT SUMMARY

A web application for small teams to create projects, manage tasks, assign owners, and view simple progress reports. The candidate will build a RESTful Spring Boot API, a MySQL data store, and an Angular front end.

CANDIDATE Varun Chopra	TYPE fullstack	DURATION 14 days
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Primary Objectives

- Implement a secure Spring Boot REST API with JWT authentication.
- Design a normalized MySQL schema for users, projects, and tasks.
- Build an Angular SPA that consumes the API and provides CRUD screens.
- Add role based access control and a simple reporting view.
- Provide automated unit and integration tests and Docker based deployment.

Candidate Details

NAME

Varun Chopra

Technology Stack

Technologies, frameworks and tools expected for implementation.

Angular Technology / Tool	TypeScript Technology / Tool	Java Technology / Tool
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Spring Boot

Technology / Tool

MySQL

Technology / Tool

REST API

Technology / Tool

Functional Requirements

- User registration with email and password validation.
- Login returning a JWT token for subsequent requests.
- Create, edit, delete projects (name, description).
- Create, edit, assign, change status, delete tasks (title, description, assignee, due date).
- Project list view with pagination and search.
- Admin report endpoint returning task counts per project.

Non-Functional Requirements

- Passwords stored with BCrypt hashing.
- API response time < 300 ms for CRUD operations.
- Input validation on both front end and back end.
- Responsive UI adhering to WCAG AA contrast guidelines.
- Codebase covered by at least 70 % automated tests.

Modules & Features

Major functional areas that make up the project.

Authentication

Handles user sign up, login, JWT issuance and validation.

- Register endpoint with email verification stub
- Login endpoint returning JWT

Project Management

CRUD operations for projects and association with owners.

- Create / edit / delete project
- List projects with pagination

Task Management

CRUD for tasks, assignment to users, status workflow.

- Create / edit / delete task
- Assign task to a project member

Reporting

Aggregates task data for admin overview.

- Endpoint returning task count per project
- Frontend chart displaying the data

Implementation Roadmap

Recommended phases and milestones for completing the project.

1

Foundation

Initialize Git repo with README, Create Spring Boot starter project, Create Angular workspace, Add Maven and npm build scripts, Configure GitHub Actions for CI

- Initialize Git repo with README
- Create Spring Boot starter project
- Create Angular workspace
- Add Maven and npm build scripts
- Configure GitHub Actions for CI

2

Core Backend

Design MySQL schema and generate JPA entities, Implement JWT based security filter, Create AuthController (register, login), Create ProjectController (CRUD), Create TaskController (CRUD)

- Design MySQL schema and generate JPA entities
- Implement JWT based security filter
- Create AuthController (register, login)
- Create ProjectController (CRUD)
- Create TaskController (CRUD)

3

Frontend UI

Set up Angular routing and guard using JWT, Create login and registration forms, Develop project list and edit components, Develop task board component, Integrate services with backend API

- Set up Angular routing and guard using JWT
- Create login and registration forms
- Develop project list and edit components
- Develop task board component
- Integrate services with backend API

4

Intermediate Features

Add ROLE_ADMIN and ROLE_USER authorities, Protect reporting endpoint with admin role, Create reporting component with chart library, Implement client side pagination and search, Add input validation feedback

- Add ROLE_ADMIN and ROLE_USER authorities
- Protect reporting endpoint with admin role
- Create reporting component with chart library
- Implement client side pagination and search
- Add input validation feedback

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Quality & Deployment

Write unit tests for services (JUnit, Mockito), Write Angular unit tests (Jest/Karma), Add integration tests for API endpoints, Configure Dockerfile for backend and frontend, Create docker compose.yml for MySQL, API, and UI

- Write unit tests for services (JUnit, Mockito)
- Write Angular unit tests (Jest/Karma)
- Add integration tests for API endpoints
- Configure Dockerfile for backend and frontend
- Create docker compose.yml for MySQL, API, and UI

API Design

Key API endpoints and expected responsibilities.

POST

/api/auth/register

Create new user account

POST

/api/auth/login

Authenticate and receive JWT

GET

/api/projects

List projects with pagination

POST**/api/projects**

Create a new project

Implementation Note

This document is generated from the project record. AI-generated requirements should be reviewed by an administrator before implementation.